

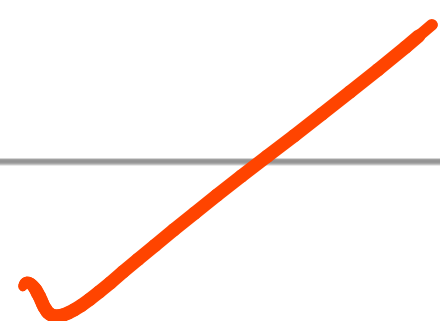
Problem 1:

$$12_{10} \& 7_{10}$$

$$= 1100_2 \& 0111_2$$

$$= 0100_2$$

$$= 4_{10}$$



Problem 2:

$$\underline{15_{10} / 10_{10}}$$

$$15_{10} \mid 10_{10}$$

$$= 1111_2 \mid 1010_2$$

$$= 1111_2$$

$$= \underline{\underline{15_2}}$$

$$15_{10}$$

$$\begin{array}{r} 1111 \\ 1 \overline{) 1010} \\ 1111 \end{array}$$

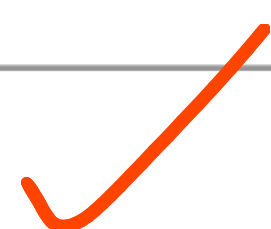
Problem 3:

$$20_{10}^{10} 13_{10}$$

$$= 10100_2^{10} 0110_2$$

$$= 1100_2$$

$$= 25_{10}$$



Problem 4:

$$\sim 10_{10} = \sim (00001010)_2 = 11110101_2$$

$$= -128 + 64 + 32 + 16 + 0 + 4 + 0 + 1$$

(2's complement form for -ve numbers)

$$= -11$$

$$= 2's \text{ complement } (10_{10})$$

$$= 2's \text{ complement } (00001010)_2$$

$$= 1's \text{ complement } (00001010)_2 + 1_2$$

$$= 1's \text{ complement } (11110101)_2 + 1_2$$

$$= 11110110_2$$

$$= -128 + 64 + 32 + 16 + 0 + 4 + 2 + 0$$

$$= -10$$

Problem 5:

$$8_{10} \ll 2$$

$$x \ll k \\ = x \times 2^k$$

k	$x \ll k$
0	x
1	$x \times 2$
2	$x \times 4$
3	$x \times 8$
$\vdots$	
k	$x \times 2^k$

$$= 1000_2 \ll 2$$

$$= 100000_2$$

$$= 32_{10}$$

Problem 6:

$$64_{10} \gg 3$$

$$64_{10} \gg 3$$

$$= 1000000_2 \gg 3$$

$$= 1000_2$$

$$= 8_{10}$$

$$= x \gg k$$

$$\neq \frac{x}{2^k}$$

k	$x \gg k$
0	x
1	

$$x = 13$$

$$k = 2$$

$$x \gg k = 13 \gg 2$$

$$33$$

$$= 1101_2 \gg 2$$

$$= 11_2 = 3$$

$$\gg 4$$

$$\frac{x}{2^k} = \frac{13}{4} = 3.25$$

$$16$$

$$100001$$

Problem 7:

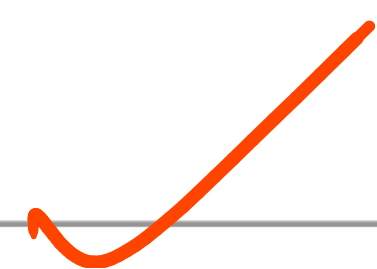
$$(5_{10} \ \& \ 3_{10}) \mid 4_{10}$$

$$= (10_2 \ \& \ 01_2) \mid 100_2$$

$$= 00_2 \mid 100_2$$

$$= 101_2$$

$$= 5_{10}$$



Problem 8:

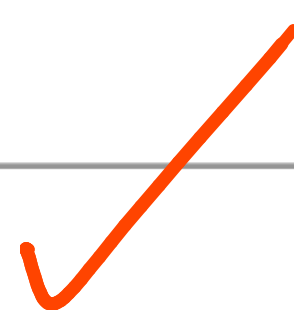
$$(10_{10} \gg 1) \& 7_{10}$$

$$= (1010_2 \gg 1) \& 111_2$$

$$= 101_2 \& 111_2$$

$$= 101_2$$

$$= 5_{10}$$



Problem 9:

$$1^1 1^1 1$$

XOR $\oplus$
is associative

$$= 1^1 (1^1 1)$$

$$= 1^1 0$$

$$= 1$$



Problem 10:

Comp: complement

$$\sim (-5_{10})$$

$$\text{not}(x) = -x - 1$$

$$\sim(-5) = -(-5) - 1 = 5 - 1 = 4$$

$$= \sim(2\text{'s comp}(5_{10}))$$

$$= \sim(2\text{'s comp}(0000010_2))$$

$$= \sim(1\text{'s comp}(0000010) + 1)$$

$$= \sim(1111101_2) = \sim(1111101_2 + 1_2)$$

$$= 00000100_2$$

$$= 4_{10}$$

Problem 11:

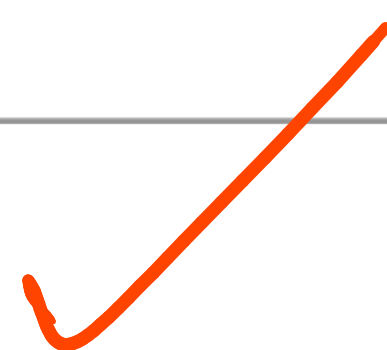
$\{ T: \text{True}, F: \text{False} \}$

$T \text{ or } F \text{ and } F$

$= T \text{ or } (F \text{ and } F)$

$= T \text{ or } F$

$= T$



Problem 12:

$\text{not}(T \text{ and } F)$

$= \text{not}(F)$ $\text{not}(F)$

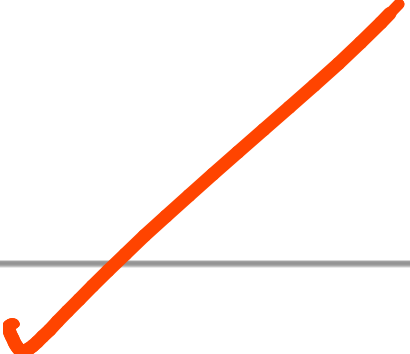
$= T$ ✓

Problem 13:

$(5 > 3)$ and $(2 == 2)$


$\Rightarrow T$ and $(2 == 2)$

$\Rightarrow T$ and T

$\Rightarrow T$


Problem 14:

0 or 5

= F or T

= T (Because in python, 0 = F and numbers ≥ 1 = T)
 < -1 are also T

(Python is giving 5. Why?)

↓
If the first operand is false for OR, result = second operand
If the first operand is true for OR, result = first operand

Problem 15:

10 and 20

= T and T

= T (Because in python, 0 =
F and numbers $\neq 0$ = T)
-1 are also T

(Python is giving 20. Why?)

For 'and', if first operand is:

(1) True ($\neq 0$) then result = second operand

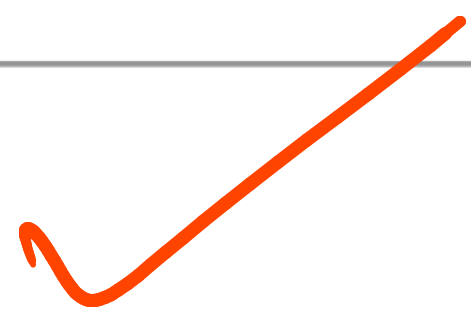
(2) False ($= 0$) then result = 0.

Problem 16:

$\text{not } ("apple" == "orange")$

$= \text{not } (F)$

$= T$

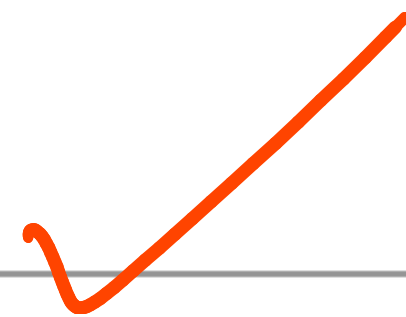


Problem 17:

$$a = T, b = T, a \neq b$$

$$a \neq b$$

$$\begin{aligned} \Rightarrow T &\neq T \\ &= F \end{aligned}$$



Problem 18:

left circular shift on 128_{10} by 1 bit

$$128_{10} \text{ circular} \ll 1$$

$$= 10000000_2 \text{ circular} \ll \text{shift}$$

$$= 0000001_2 = 1$$

$$= 01000000_2$$

$$= 64_{10}$$

X

X

$$n = 00abcdef_2$$

$$n \ll 6$$

$$A = (abcdef00000000_2) \vee (100000000_2)$$

$$= ef000000$$

$$B = 00abcdef_2 \gg 2 = 00abcd_2$$

$$A \vee B = ef00abcd_2$$

Eg: $101_2 \ll 6$ bits

$$N = 8 \text{ (total bits)}$$

$$(00000101_2) \ll 6$$

rotated
A left part = (00000101000000_2)

$$\% 10000000$$

$$= 01000000_2$$

(last 8 bits)

rotated
B right part = $(n \gg (N-d))$

$$= 101 \gg (8-6)$$

$$= 101_2 \gg 2 = 1_2$$

$$\text{ans} = A \vee B = 01000001$$

Problem 19:

$$\text{bool}(C10)^\wedge \text{bool}(C0)$$

$$= T^\wedge \text{bool}(C0)$$

$$= T^\wedge F$$

$$= T$$



Problem 20:

$$(7_{10} \& 1) == 1 \text{ and } (8_{10} \gg 1) == 4_{10}$$

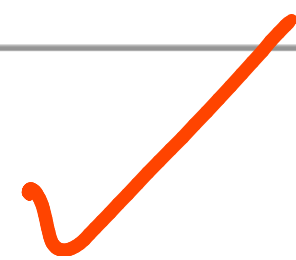
$$\Rightarrow (111_2 \& 1) == 1 \text{ and } (1000_2 \gg 1) == 100_2$$

$111_2 \& 001_2$

$$\Rightarrow 1 == 1 \text{ and } (1000_2 \gg 1) == 100_2$$

$= T \text{ and } T$

$= T$



Problem 21:

$T \text{ or } F \text{ and not } T$

$\equiv T \text{ or } (F \text{ and not } T)$

$\equiv T \text{ or } (F \text{ and } F)$

$\equiv T \text{ or } F$



$\equiv T$

Prob (em 22:

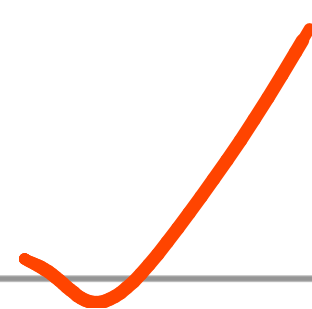
$(23 > 42) \text{ and } (36 > 12) \text{ or } (39 \leq 39)$

$= F \text{ and } (36 > 12) \text{ or } (39 \leq 39)$

$= F \text{ and } T \text{ or } (39 \leq 39)$

$= F \text{ and } T \text{ or } T$

$= F \text{ or } T$



$= T$

Problem 23:

$$10_{10} + 5_{10} << 1$$

$$= (10 + 5)_{10} << 1$$

$$= 15_{10} << 1$$

$$= 111_2 << 1$$

$$= 1110_2$$

$$= 30_{10}$$

Problem 24:

$A \text{ or } (\text{not } A \text{ and } B)$

$= (A \text{ or not } A) \text{ and } (A \text{ or } B)$

$= \text{True and } (A \text{ or } B)$

$= A \text{ or } B$

Problem 25:

$(A \text{ and } B) \text{ or } (A \text{ and not } B)$

$= A \text{ and } (B \text{ or not } B)$

$= A \text{ and } T$

$= A$

Problem 26:

$A \text{ or } CF \text{ and } B)$

$$= A \text{ or } F$$

$$= A$$

Problem 27:

$\text{not}(\text{not } A \text{ or } B)$

~~$= (\text{not}(\text{not } A) \text{ and } B)$~~

$= (\text{not}(\text{not } A)) \text{ and } (\text{not } B)$

$= A \text{ and not } B$

Problem 28:

$$5_{10} \times 2_{10}^2 \& 15_{10}$$

$$= (5_{10} \times (2_{10}^2)) \& 15_{10}$$

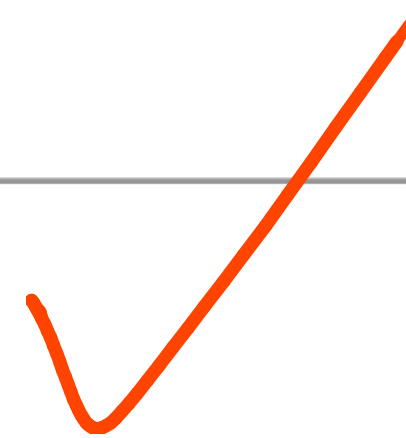
$$= (5_{10} \times 4_{10}) \& 15_{10}$$

$$= 20_{10} \& 15_{10}$$

$$= 10100_2 \& 01111_2$$

$$= 00100_2$$

$$= 4_{10}$$



Problem 29:

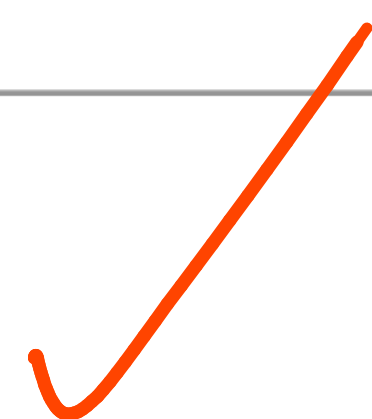
$$\text{not } (15 // 3 == 5 \text{ and } 7 / 2 == 0)$$

$$= \text{not } (5 == 5 \text{ and } 1 == 0)$$

$$= \text{not } (T \text{ and } F)$$

$$= \text{not } (F)$$

$$= T$$



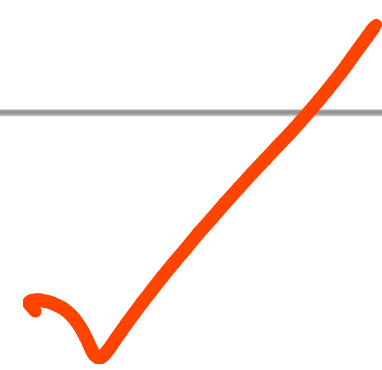
Problem 30:

$$(A \text{ or } B) \text{ and } (\text{not } A \text{ or } B)$$

$$= (A \text{ and not } A) \text{ or } B$$

$$= F \text{ or } B$$

$$= B$$



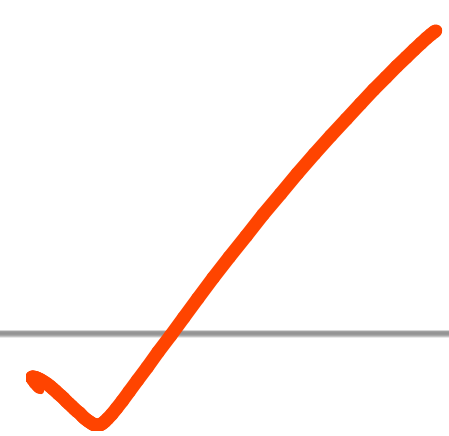
Problem 31:

$(A \text{ and } B \text{ and } C) \text{ or } (A \text{ and } B \text{ and not } C)$

$= A \text{ and } B \text{ and } (C \text{ or not } C)$

$= A \text{ and } B \text{ and } T$

$= A \text{ and } B$



Problem 32:

$$\sim 5_{10} \& 7$$

$$= \sim 10_2 \& 111_2$$

$$= 010_2 \& 111_2$$

$$= 10_2$$

$$= 2$$

$$= \sim (00000101_2) \& (111_2)$$

$$= (\underline{1111}010)_2 \& (\underline{0000}111_2)$$

$$= (0000010)_2$$

$$= 2$$

$$\sim 5_{10} \mid 7_{10} \quad \text{with 2's complement}$$

$$111_2$$

$$= 7$$

$$(1111010)_2 \mid (0000111)_2$$

$$= 1111111_2$$

$$= -128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

$$= -1$$

Problem 33:

$$|cc3| |<<1$$

$$1000_2 | 0010_2$$

$$= 10|0_2$$

$$= 10_{10}$$

Problem 34:

A and (a or b)

= a (absorption)

A

Problem 35:

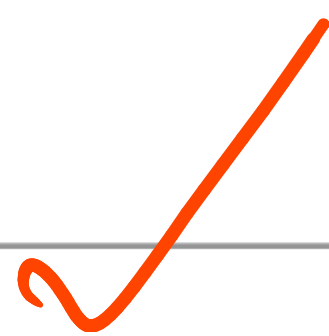
$(T \text{ or } F) \text{ and } (\text{not } T \text{ or } T)$

$= T \text{ and } (\text{not } T \text{ or } T)$

$= T \text{ and } (F \text{ or } T)$

$= T \text{ and } T$

$= T$



Problem 36.

$$A \text{ or } (A \text{ and } B) \text{ or } (B \text{ or } F)$$

$$= A \text{ or } (B \text{ or } F) \quad (\text{Absorption})$$

$$= A \text{ or } \text{F}$$

$$= A \text{ or } F$$

$$= A$$

problem 37:

1_{10} 8-bit circular $\gg 1$

$= 00000001_2$ circular $\gg 1$

$= 10000000_2$

$= 256_{10}$

128_{10}

X

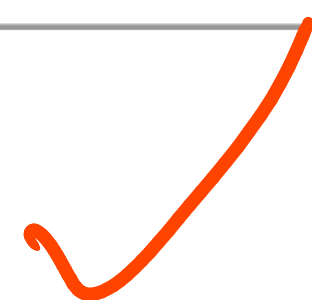
Problem 38:

$$10 > 5 > 2$$

$$= (10 > 5) \text{ and } (5 > 2)$$

$$= T \text{ and } T$$

$$= T$$



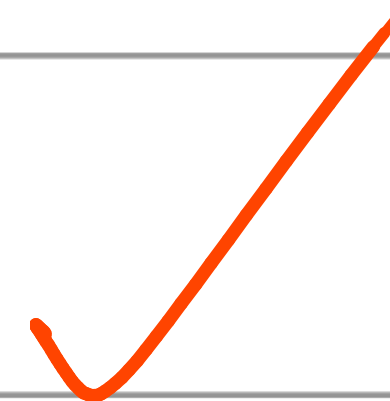
Problem 39:

$$\text{not}(5 < 2) \text{ or } 10 // 2 == 5$$

$$= \text{not}(f) \text{ or } 10 // 2 == 5$$

$$= T \text{ or } 10 // 2 == 5$$

$$= T \text{ or } 5 == 5$$



$$= T \text{ or } T$$

$$= T$$

Problem 40:

$$(A \text{ and } B) \text{ or } C = A \text{ or } C \text{ \& } B \text{ or } C$$

$$(A \text{ and } B) \text{ or } (\text{not } (A \text{ and } B))$$

$$= (A \text{ and } B) \text{ or } (\text{not } A \text{ or } \text{not } B)$$

$$= ((A \text{ and } B) \text{ or } \text{not } A) \text{ or } \text{not } B$$

$$= (\underline{(A \text{ or } \text{not } A) \text{ and } (B \text{ or } \text{not } A)}) \text{ or } \text{not } B$$

$$= (T) \text{ and } (B \text{ or } \text{not } A) \text{ or } \text{not } B$$

$$= (B \text{ or } \text{not } A) \text{ or } \text{not } B$$

✓

$$= (B \text{ or } \text{not } A) \text{ or } \text{not } B$$

$$= (B \text{ or } \text{not } B) \text{ or } (\text{not } A) \quad [\text{associative}]$$

$$= T \text{ or } \text{not } A$$

$$= T$$

Let A and $B = X$.

$$(A \text{ and } B) \text{ or } (\text{not } (A \text{ and } B))$$

$$= X \text{ or } \text{not } X$$

$$= T$$